

NANYANG TECHNOLOGICAL UNIVERSITY

School of Social Sciences

HE3002 Macroeconomics III

LECTURE 7

The IS-MPR-PC Model II

THE IS CURVE

1. An IS curve relates output to interest rates.
2. The higher interest rates are, the lower output is.
3. It should be noted that the IS relationship is between output and **real** interest rates, not **nominal** rates.
4. Real interest rates adjust the headline (nominal) interest rate by subtracting off inflation i.e.

$$r_t = i_t - \pi_t \quad (1)$$

where r_t is the real interest rate, i_t is the nominal interest rate and π_t is the rate of inflation.

5. The IS curve has the following form:

$$y_t = y_t^* - \alpha(i_t - \pi_t - r^*) + \epsilon_t^y \quad (2)$$

6. Expressed in words, this equation states that the gap between output and its natural rate ($y_t - y_t^*$) depends on two factors:

- (a) **The real interest rate $i_t - \pi_t$:** The coefficient α describes the effect of a one point increase in the real interest rate on output. We can see from the equation that if $\epsilon_t^y = 0$ then a real interest rate of r^* will imply $y_t = y_t^*$.
- (b) **Aggregate demand shocks, ϵ_t^y :** Many other factors beyond the real interest rate influence aggregate spending decisions. These include fiscal policy, asset prices and consumer and business sentiment. We will model all of these factors as temporary deviations from zero of an aggregate demand shock, ϵ_t^y .

A MONETARY POLICY RULE

1. We will consider the following monetary policy rules that the central bank may follow in our model:

$$i_t = r^* + \pi^* + \beta_\pi(\pi_t - \pi^*) \quad (3)$$

2. The central bank adjusts its interest rate in line with inflation with the goal of meeting an inflation target.
3. Specifically, the central bank adjusts the nominal interest rate, i_t , upwards when inflation, π_t , goes up and downwards when inflation goes down (we are assuming that $\beta_\pi > 0$) and it does so in a way that means when inflation equals a target level, π^* , chosen by the central bank, real interest rates will be equal to their natural level.
4. Why a rule of this form might be a good idea?
5. Suppose the central bank has a target inflation rate of π^* that it wants to achieve. Ideally, it would like the public to understand that this is the likely inflation rate that will occur, so that $\pi_t^e = \pi^*$. If that can be achieved, then the Phillips curve tells us that, on average, inflation will equal π^* provided we have $y_t = y_t^*$. And the IS curve tells us that, on average, we will have $y_t = y_t^*$ when $i_t - \pi_t = r^*$. So this is a policy that can help to enforce an average inflation rate equal to the central bank's desired target, provided the public adjusts its inflationary expectations accordingly.

THE IS-MPR CURVE

1. This is the model. It consists of three equations: (a) the Phillips curve, (b) the IS curve and (c) the Monetary Policy Rule.
2. Specifically,

(a) The Phillips curve:

$$\pi_t = \pi_t^e + \gamma(y_t - y_t^*) + \epsilon_t^\pi \quad (4)$$

(b) The IS curve:

$$y_t = y_t^* - \alpha(i_t - \pi_t - r^*) + \epsilon_t^y \quad (5)$$

(c) The Monetary Policy Rule:

$$i_t = r^* + \pi^* + \beta_\pi(\pi_t - \pi^*) \quad (6)$$

3. What about a graphical representation of this model?
4. This is a system of **three** variables which makes it hard to express on a graph with two axes.
5. To make the model easier to analyse using graphs, we are going to reduce it down to a system with two main variables (inflation and output).
6. We can do this because the monetary policy rule makes interest rates a function of inflation, so we can substitute this rule into the IS curve to get a new relationship between output and inflation that we will call the IS-MPR curve.
7. Substituting the MPR equation (10) into the IS equation (9) gives:

$$y_t = y_t^* - \alpha(\beta_\pi - 1)(\pi_t - \pi^*) + \epsilon_t^y \quad (7)$$

8. This is the IS-MPR curve. It combines the information in the IS curve and the MRP curve into one relationship.
9. What would this curve look like?
10. It turns out it depends especially on the value of β_π , the parameter that describes how the central bank reacts to inflation.
11. The IS-MPR curve says that an extra unit of inflation implies a change of $-\alpha(\beta_\pi - 1)$ in output.
12. Is this positive or negative?

13. We are assuming that $\alpha > 0$ so this combined coefficient will be negative if $\beta_\pi - 1 > 0$.
14. In other words, the IS-MPR curve will have a negative slope provided the central bank reacts to inflation so that $\beta_\pi > 1$.
15. The explanation for this is as follows: An increase in inflation of x will lead to an increase in nominal interest rates of $\beta_\pi x$ so real interest rates change by $(\beta_\pi - 1)x$. This means that if $\beta_\pi > 1$, an increase in inflation leads to higher real interest rates and, via the IS curve relation, to lower output. So if $\beta_\pi > 1$ then the IS-MPR curve can be depicted as a downward-sloping curve.
16. In contrast, if $\beta_\pi < 1$, then an increase in inflation leads to lower real interest rates and higher output, implying an upward-sloping IS-MPR curve.
17. For now, we assume that $\beta_\pi > 1$ so that we have a downward sloping IS-MPR curve.

[Figure 7: The IS-MPR curve with $\epsilon_t^y = 0$]

18. Note that when $\epsilon_t^y > 0$, the economy experiences an aggregate demand shock and the IS-MPR curve will shift to the right.

[**Figure 8:** The IS-MPR curve with $\epsilon_t^y > 0$]

19. Figure 7 thus shows what the IS-MPR curve looks like when the aggregate demand shock $\epsilon_t^y = 0$.
20. Again, the key thing to notice is the value of inflation that occurs when output equals its natural rate. When $y_t = y_t^*$ we get $\pi_t = \pi^*$.
21. Figure 8 shows how the IS-MPR curve shifts to the right if $\epsilon_t^y > 0$ corresponding to a positive shock to aggregate demand.

THE IS-MPR-PC MODEL

1. The full IS-MPR-PC model can be illustrated in the traditional fashion as a graph with one curve that slopes upwards (the Phillips curve) and one that slopes downwards (the IS-MPR curve provided that we assume $\beta_\pi > 1$.)

[**Figure 9:** The IS-MPR-PC curve with $\epsilon_t^\pi = 0$ and $\epsilon_t^y = 0$]

2. Figure 9 provides the simplest possible example of the graph. This is the case where both the temporary shocks, ϵ_t^π and ϵ_t^y equal zero and the public's expectation of inflation is equal to the central bank's inflation target.
3. In the next lecture, we will analyse this model in depth, examining what happens when various types of events occur and focusing carefully on how inflation expectations change over time.

INFLATION EXPECTATIONS AND INFLATION TARGET

1. We will start by repeating a graph from the last time.

[**Figure 10:** The IS-MPR-PC model when expected inflation equals the inflation target]

2. Figure 10 shows the case where both the temporary shocks, $\epsilon_t^\pi = 0$ and $\epsilon_t^y = 0$ and the public's expectation of inflation equals the central bank's inflation target.
3. Specifically, the graph shows a case where the public's expectation of inflation and the central bank's inflation target are the same $\pi_t^e = \pi_0 = \pi^* = \pi_0$.
4. With no temporary shocks, the value of output is consistent with $\pi_t = \pi_1$ for the IS-MPR curve is y^* and $y_t = y_t^*$.
5. Now consider a case in which the public's inflation expectations shift to being higher than the central bank's target rate. Figure 11 illustrates this case. It shows the PC curve shifting upwards.

[**Figure 11:** The IS-MPR-PC model when expected inflation is higher]

6. This new PC is determined by the new higher level of expected inflation. Specifically, the public's inflation expectations are now determined by $\pi_t^e = \pi_1$
7. The outcomes for inflation and output of the increase in inflation expectations are described by the intersection of the new PC and the old unchanged IS-MPR.
8. The actual outcome for inflation ends up being higher than the central bank's inflation target but lower than the public's inflationary expectations. Output ends up being lower than its natural rate because the higher level of inflation leads the central bank to raise real interest rates which reduces output.
9. Can We Learn More?
10. Figure 2 is a good example of how we can use graphs to illustrate a model's properties. It gets the basic story across as to what happens when inflation expectations rise above target when the central bank is pursuing a monetary policy rule that increases real rates in response to higher inflation.
11. Still, one could look to dig a bit deeper. The inflation outcome is slightly more than halfway towards the public's inflation expectations relative to the central bank's inflation target.

12. But what actually determines this outcome?
13. In other words, what determines how far from away target inflation will move when the public's inflation expectations change?
14. How much does it depend on the monetary policy rule?
15. How much does it depend on other aspects of the model, like the impact of real interest rates on output and the impact of output on inflation?

THE IS-MPR-PC MODEL SOLUTION FOR INFLATION

1. Let's repeat the equations describing our two curves as presented in our last set of notes. The PC curve is:

$$\pi_t = \pi_t^e + \gamma(y_t - y_t^*) + \epsilon_t^\pi \quad (8)$$

2. And the IS-MP curve is:

$$y_t = y_t^* - \alpha(\beta_\pi - 1)(\pi_t - \pi^*) + \epsilon_t^y$$

3. Taking all the other elements of the model as given, we can view this as two linear equations in the two variables y_t and π_t . These equations can be solved to give solutions that describe how these two variables depend on all the other elements of the model.
4. Accordingly, we solve for π_t as:

$$\pi_t = \left(\frac{1}{1 + \alpha\gamma(\beta_\pi - 1)} \right) \pi_t^e + \left(\frac{\alpha\gamma(\beta_\pi - 1)}{1 + \alpha\gamma(\beta_\pi - 1)} \right) \pi_t^* + \frac{\gamma\epsilon_t^y + \epsilon_t^\pi}{1 + \alpha\gamma(\beta_\pi - 1)} \quad (9)$$

5. We define:

$$\theta = \frac{1}{1 + \alpha\gamma(\beta_\pi - 1)} \quad (10)$$

so that we can re-write equation (2) for inflation as:

$$\pi_t = \theta\pi_t^e + (1 - \theta)\pi^* + \theta(\gamma\epsilon_t^y + \epsilon_t^\pi) \quad (11)$$

6. This equation shows that, apart from the shocks to output and inflation ($\theta(\gamma\epsilon_t^y + \epsilon_t^\pi)$), inflation is a weighted average of the public's inflation expectations and the central bank's inflation target i.e. it must lie between these two values as long as $0 < \theta < 1$ (which it should be).
7. What determines whether inflation depends more on the public's expectations or the central bank's target?

8. In other words, what determines the value of θ ?
9. Three different factors determine this value:
 - (a) γ : This is the parameter that determines how inflation changes when output changes. The central bank can only influence inflation by influencing output. If the effect of output on inflation gets bigger, then the central bank's inflation target will have more influence on outcome for inflation.
 - (b) α : This is the parameter that determines how output changes when output interest rates. If the effect of interest rates on output gets bigger, then the central bank's inflation target will have more influence on the outcome for inflation.
 - (c) β_π : Let's continue to assume $\beta_\pi > 1$. Then as β_π gets bigger, the central bank is reacting more to inflation being above its target level. So this parameter getting bigger means less weight on in inflation expectations in determining the outcome for inflation and more weight on the central bank's inflation target.

THE IS-MPR-PC MODEL SOLUTION FOR OUTPUT

1. Next we provide an expression for output and simplify the solution to output as:

$$y_t = y_t^* - \theta\alpha(\beta_\pi - 1)(\pi_t^e - \pi^* + \epsilon_t^\pi) + (1 - \theta\alpha\gamma(\beta_\pi - 1))\epsilon_t^y \quad (12)$$

2. This equation tells us that whether output is above or below target depends upon the gap between expected inflation and the inflation target as well as on the two temporary shocks ϵ_t^y and ϵ_t^π .
3. Provided we have the usual condition that $\beta_\pi > 1$, the combined coefficient $-\theta\alpha(\beta_\pi - 1)$ is negative. This means that:
 - (a) Increases in the public's inflation expectations relative to the inflation target end up having a negative effect on output.
 - (b) Inflationary supply shocks ($\epsilon_t^\pi > 0$) also have a negative effect on output while positive aggregate demand shocks ($\epsilon_t^y > 0$) have a positive effect on output.
4. How far does output fall short of its natural rate when inflation expectations rise above the central bank's target?
5. The coefficient determining this is $-\theta\alpha(\beta_\pi - 1)$. This can be re-expressed as:

$$-\theta\alpha(\beta_\pi - 1) = \frac{\alpha(\beta_\pi - 1)}{1 + \alpha\gamma(\beta_\pi - 1)}$$

6. Calculating partial derivatives, you find that the size of the short fall in output depends positively on α and β_π .
7. In other words, the larger the impact of interest rates on output and the larger the central bank's interest rate response to inflation, the larger the shortfall in output will be when inflation expectations rise above the central bank's target.
8. In contrast, the output shortfall depends negatively on γ , the parameter determining the effect of output on inflation: As γ gets bigger, the central bank requires a smaller shortfall in output to implement its policy of getting inflation back to target.